Reference Event 3: Ukraine's Payment System Through the War, 2022 to 2023

Characterization Against the Financial Services Sector Design Basis

DBA-FIN-RE-3 — Reference Event Instrument, DBA-FIN Series

Lineage: Issued under the Financial Services Sector Framework v0.5; DBA-MA Parent Framework; conferred on the proposal of Payment and Settlement (DBA-FIN-PS-FC1)

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This version is a member of CIKR Design Basis Examples, Edition 1 (November 2026) (https://doi.org/10.5281/zenodo.23863986), a dated edition that freezes, for each of fifteen critical infrastructure sectors, a sector framework, a facility class and a reference event as they stood on 1 November 2026. The edition is not falsified by a later revision of this document, whose working master continues to move. It is a working draft, marked DRAFT on its cover.

1. Purpose, Scope and Position

1.1 Purpose

This instrument reads the operation of Ukraine's payment system through the war back through DBA-FIN-PS-FC1 (*Payment and Settlement*), whose governing military-action scenario, settlement under a campaign on the grid and communications, it characterizes (§6.3), at the class's first reference position, the central-bank real-time gross settlement system (§1.3), and at the participants and the public who must reach it. It asks whether the class's design basis anticipated what happened, and returns what it finds.

1.2 Position and Scope

A reference event instrument, in the series convention, reads a real event backward through the design basis, adds no requirements of its own, and returns findings and recommendations for disposition. This one is issued under the Financial Services Sector Framework v0.5, inherits the DBA-MA parent framework, and was conferred by the framework's §9.1 on the author's decision of 14 October 2026, on the proposal of *Payment and Settlement*, as a continuing condition for this instrument to bound. Section references of the bare form §N are to that class at v0.1, the version this instrument read.

The event is a condition, not a day, and this instrument bounds it so. It begins on 24 February 2022, with the full-scale invasion. Its second phase begins on 10 October 2022, the first day of the campaign of missile strikes on the electricity system that the International Criminal Court's warrants cover. The reading closes at 28 July 2023, the date of the National Bank of Ukraine's principal published account; the attacks on generation of 2024 are recorded only as showing the condition continued. The condition continues, and the record is open.

No individual is named; the National Bank's officers are named by role, and the National Bank, the court and the United Nations mission as institutions. State attribution is each government's own and each court's and mission's finding is its own: the attacks on the electricity system are attributed to the Russian armed forces by the court's warrants and the mission's reports, and the denial-of-service attacks on Ukrainian banks of February 2022 to Russia's military intelligence service by the governments of the United States and the United Kingdom. This instrument makes no finding of attribution and none of fault.

The System of Electronic Payments, the banks' access to it and the public's access to payment lie within this class; the banks' customer services and online banking are the core-banking class's; the grid is the energy series'. The Foundational Definitions are cited by designation, FD-REG holding their versions: FD-DBL (*Design Basis Layers*) bears on Section 5.

2. Sources and Attribution

2.1 Sources and Their Limit

The National Bank’s own accounts are primary: its release on the System of Electronic Payments in 2022; its releases on the working Saturday of 12 March 2022, on the Power Banking network and on the new generation of the system launched on 1 April 2023; an article by its deputy governor for payment and settlement systems of 13 May 2022; and the interview its director for payment systems gave the European Payments Council, published on 28 July 2023. The campaign is read from the International Criminal Court’s announcement of 5 March 2024 and the United Nations Human Rights Monitoring Mission in Ukraine’s release of 19 September 2024; the attacks on banks from the statements of the United States and United Kingdom governments of 18 February 2022. The National Bank’s annual report for 2022 could not be retrieved, and its account of the system’s resilience arrangements is taken from the releases and the interview. No non-public material is used.

2.2 Where the Sources Differ

The National Bank’s release for 2022 gives the system’s share of hryvnia interbank transfers as about 97 percent; the interview of 2023 gives more than 98 percent. The Power Banking network was announced with more than 1,000 branches and had more than 2,370 by July 2023. Each figure is stated with its date and author.

3. The Event

3.1 The System and the Preparation

The System of Electronic Payments is the National Bank’s real-time gross settlement system for hryvnia. The National Bank and the banks drew up contingency plans in 2014, after the annexation of Crimea, and activated them when the full-scale invasion began (interview).

3.2 The First Phase, From 24 February 2022

The National Bank states that the system has operated as usual and without disruptions since the first day of the invasion. It kept the system operating twenty-four hours a day, deploying its backup on reserve sites (deputy governor, May 2022); in March 2022 the system ran twenty-three hours a day and Saturday 12 March was made a banking business day. In the first weeks of the war the National Bank authorized banks to process and store information in cloud services hosted in the European Union, the United Kingdom, the United States and Canada, so that they could resume if their facilities in Ukraine were destroyed. In 2022 the system processed about 363 million payments worth over 133 trillion hryvnia, about 1.4 million a banking day, with no disruption reported. Card payments carried on without restriction, and reached about 93 percent of transactions in May to December 2022 (release for 2022; interview).

3.3 The Attacks on Banks

On 15 and 16 February 2022 denial-of-service attacks took offline the websites of the country's largest bank and its state savings bank, among other targets. The United States government stated on 18 February that it had technical information linking Russia's military intelligence service to the attacks, and the United Kingdom government judged the service involved. The attacks reached banks' online services, which are the core-banking class's; no effect on the System of Electronic Payments is recorded.

3.4 The Second Phase, From 10 October 2022

From 10 October 2022 the Russian armed forces conducted, on the International Criminal Court's finding of reasonable grounds to believe, a campaign of missile strikes on electric power plants and substations that ran at least to 9 March 2023; the court issued two warrants on 5 March 2024 for directing attacks at civilian objects and for the crime against humanity of other inhumane acts. The National Bank's director for payment systems describes the strikes as aiming to plunge the country into total blackout. The United Nations mission records nine further waves of coordinated attacks between March and August 2024, now directed to a much larger extent at generation, with about nine gigawatts of capacity destroyed and outages of twelve hours or more a day in some cities.

3.5 The Response at the Edge

The National Bank and the banks formed the Power Banking network: branches with alternative energy sources, backup communication channels, enhanced cash collection and additional staff, so that the public could withdraw cash, make payments and transfers and exchange currency through blackouts. All fourteen systemically important banks form its basis. It was announced with more than 1,000 branches and had more than 2,370 in about 390 cities and towns by July 2023 (National Bank release; interview).

3.6 The Change During the War

On 1 April 2023 the National Bank launched a new generation of the system, on the ISO 20022 standard, open for interbank payments twenty-four hours a day, seven days a week.

4. Technical Decomposition

The condition decomposes into initiating elements (I-1 to I-4) and effect elements (E-1 to E-4), as Table 4-1 sets out. Composed, they are the class's governing military-action scenario (§6.3) at the central-bank system and its participants (§1.3), with Chain F postulated and not realized, through a condition of years.

Table 4-1. Event elements and their mapping. Section references are to *Payment and Settlement* v0.1.

No.	What happened	Class provision	Section
I-1	Full-scale invasion; plans of 2014 activated	Applicability determination (§6.1)	3.1, 3.2
I-2	Denial-of-service attacks on the largest banks, attributed by two governments	Cross-sector interaction (§6.6)	3.3
I-3	Campaign of strikes on power plants and substations from October 2022	Governing scenario (§6.3)	3.4
I-4	Strikes shift to generation in 2024; outages of twelve hours or more	Endurance (§6.8, second residual)	3.4
E-1	System operates without disruption; backup on reserve sites; banks' data abroad	Sites not sharing the exposure (§6.3, §6.7)	3.2
E-2	Hours extended to 23 and a working Saturday; then 24/7 from April 2023	Extension capacity (§3.3)	3.2, 3.6
E-3	Power Banking branches with their own power and communications	The public's access (§6.3)	3.5
E-4	No alteration of the system's records recorded	Integrity at each resumption (§6.3, §6.5)	3.2

5. Tier Determination

5.1 The Four Characteristics

The characteristics are read against the condition the class's §6.3 postulates, a campaign on the grid and communications on which settlement depends, in the forms *Payment and Settlement* §6.2 states. Intentionality is present: the campaign on the electricity system was deliberate on the court's and the mission's findings, and the attacks on the largest banks of February 2022 were deliberate on the two governments' statements. Persistence is present: the campaign ran from October 2022 to at least March 2023 and recurred in nine waves in 2024, through more than three years of war. Precision is present: the strikes were directed at power plants and substations, and in 2024 at generation, whose loss produces the longest recovery, and the attacks of February 2022 were directed at the country's largest bank and its state savings bank. Denial is present: the strikes removed the power on which the system's participants and the public's access depend, for hours a day, through winters. The record does not show that the System of Electronic Payments itself was an aim of the campaign, and this instrument does not infer it.

5.2 Determination

The event meets the threshold for the military-action extension. All four characteristics are established on the record, as FD-DBL (*Design Basis Layers*) §6.4 requires under the author's rulings of 10 and 13 October 2026, in the forms the class states, and the actor is a state's armed forces on the court's and the mission's findings. It is a military-action event of the class's governing scenario (§6.3), in which Chain F was postulated and did not occur, at the reach of a

country and a duration of years. It is the series' first reference event that records an answer: on the National Bank's account, the Assured Finality Condition of §6.7 held at the center, and access held at the edge by provision the class does not yet require. That is the tier determination returned for the DBA Reference Event Register. Energy is proposed as an impacted sector, as the sector whose denial the class withstood.

6. Characterization Against the Design Basis

6.1 §6.3 The Class's Account of the Event

Held, with a gap. The account holds in substance: the plans of 2014, the cloud services abroad, the system working as reliably as before, the strikes from autumn 2022 and the branches with their own power. Two points are qualified and go to the class as Erratum 2 of Project 11's note of 14 October 2026 (Financial Services batch): the share of interbank payments should be given as the National Bank states it in each source, about 97 percent in its release for 2022 and more than 98 percent in the interview of 2023; and the 2,370 branches are the network as of July 2023, announced with more than 1,000. The conferral note asked that the National Bank's own reports be sought; its releases and its deputy governor's article are now read and confirm the account, and its annual report could not be retrieved (Section 2.1).

6.2 §6.3 and §6.7 Settlement Through the Campaign

Held. The system settled every banking day through the campaign, from reserve sites, with no disruption reported, and the banks' data and operations could resume from cloud services outside the country. The class's requirement that the system settle from sites and arrangements that do not share the exposure the campaign targets is the event.

6.3 §6.3 and §6.7 The Public's Access to Payment

Held, with a gap. The class requires the system to settle and its participants to reach it, and its design-basis reading records that access was kept at the edge. Its requirements do not reach the edge: the branches, cash and card acceptance through which the public pays and is paid when the grid fails. The National Bank found it had to build that edge in the second phase. Finding F-1.

6.4 §6.1 and §6.3 Preparation and the Permissions It Needs

Held, with a gap. The plans of 2014 were activated in 2022, as the class's applicability determination contemplates. The arrangement that let banks resume from outside the country rested on a permission the regulator granted in the first weeks of the war, after the invasion had begun. Finding F-2.

6.5 §3.3 The Extension Capacity

Held. The system's hours were extended to twenty-three a day and a Saturday made a business day in March 2022, and from April 2023 the system ran at all hours. The class's extension capacity, used as an instrument of continuity rather than of recovery, held.

6.6 §7.2 A Change of Generation During the Campaign

Held. The National Bank replaced the system with a new generation in April 2023, in wartime, and no disruption is reported. The sources read do not say how the change was tested; the class's posture, that each configuration a change produces is tested before it takes effect, is not contradicted.

6.7 §6.5 and §5.2 The Integrity of the Records

Not exercised. No alteration of the system's records is recorded in the sources read. The attacks on banks of February 2022 reached their online services, which are the core-banking class's.

6.8 §6.8 The Second Declared Residual

Held. The class declares as a residual a campaign that outlasts the endurance of the system's own power, communications and time. Through the period read, endurance held; the residual is not realized, and the shift of the strikes to generation in 2024 keeps it in view.

6.9 §6.4 Exclusion as Denial

Not exercised. No system or network was closed to Ukraine's participants in the sources read.

6.10 §7.3 The AI-Governance Design Basis

Not exercised. No component within the scope of *Layer Decomposition and Determinism* §4.3 bore on the event in the sources read.

6.11 Summary of Design Basis Performance

Of the ten readings, three Held, with a gap, the class's account among them; four Held; and three were Not exercised. Nothing was Missed. The class's governing scenario was written with this event as its case, and the event is the scenario carried out, with the answer in place on the National Bank's account. What the event adds is that the answer reached the public only because the edge was built, and that the arrangements for operating from outside the theater rested on permissions granted after the war began.

7. Findings

Section references are to *Payment and Settlement* v0.1 unless another instrument is named.

7.1 F-1 — The Assured Finality Condition Reaches the Edge

When strikes on the grid began in the autumn of 2022, the National Bank and the banks formed a network of branches with alternative energy, backup communications and enhanced cash collection, more than 2,370 by July 2023, so that the public could withdraw cash, pay and transfer through blackouts; card payments carried on and rose to about 93 percent of transactions (Sections 3.2 and 3.5). The finding confirms §6.3's statement that the public must be able to pay and be paid, and is that the class should state, as part of its governing scenario and of the Assured Finality Condition of §6.7, a requirement on second-tier participants together: that enough points of access, with power, communications and cash of their own for the campaign's longest outage, are held across the country, their number and distribution stated in advance; and that the retail revision of §10.3 takes this up for the card networks and instant payment systems.

Affected: §6.3, §6.7, §10.3. Disposition: R-1; noted to Core Banking and Ledger.

7.2 F-2 — The Permissions Continuity Needs Are Granted Before the War

The plans of 2014 were activated in February 2022; the authorization for banks to process and store their information in cloud services abroad, so that they could resume if their facilities were destroyed, was granted in the first weeks of the war (Sections 3.1 and 3.2). The finding confirms §6.3's requirement of sites and arrangements outside the exposure of the campaign, and is that the class should state that, where a system or participant has a positive applicability determination under §6.1, the regulatory permissions its continuity arrangements depend on (where data may be held, which services may be used from abroad, which days and hours may be declared for settlement) are granted or pre-authorized, with their conditions, before the conflict, and are exercised.

Affected: §6.1, §6.3, §8.4. Disposition: R-2.

8. Recommendation Register

Priority is on the series' one-to-three scale, 1 the most consequential. R-1 and R-2 dispose to *Payment and Settlement*.

No.	Title	Affected	Recommendation	Pri.
R-1	Carry the Assured Finality Condition to the edge	§6.3, §6.7, §10.3	Require enough access points, with their own power, communications and cash for the campaign's longest outage, their number and distribution stated in advance; carry to the retail revision. Noted to Core Banking and Ledger.	1
R-2	Grant continuity permissions before the war	§6.1, §6.3, §8.4	Where applicability is positive, grant or pre-authorize, with conditions, the permissions continuity depends on (data location, services from abroad, settlement days and hours) and exercise	2

			them.	

9. Open Items

OI-1: the register row. Project 0 is asked to enter the tier determination of Section 5.2 and the impacted sector. OI-2: the class has not taken in this instrument and conforms at its next revision, with Erratum 2. OI-3: the condition continues; the instrument is revised if the record after 28 July 2023 shows the system disrupted or its records altered, and its annual report is read when it can be retrieved. OI-4: closed. The author ruled on 14 October 2026 that the four characteristics may be read against the campaign on a class's dependencies, rather than against the facility as an aim, as this instrument and *Air-Raid Alerting in Ukraine* (DBA-ESS-RE-9) §5 read them.

10. Standing of This Document

10.1 What This Instrument Establishes

It modifies neither class nor framework, and makes no finding of attribution and none of fault.

10.2 Relationship to the Class's Use Case

The class's use case traces §6.3. This event is that scenario carried out, so the use case may run on its figures rather than construct them: a system settling about 1.4 million payments a banking day without disruption from reserve sites; plans prepared eight years before; banks' data abroad by permission granted in weeks; hours extended to twenty-three and then to all hours; outages of twelve hours or more a day; and an edge of more than 2,370 branches with their own power. F-1 and F-2 bear most on the use case. What the use case adds, an adversary who also alters the records (§6.5), is its own construction; the event records none.

10.3 Relationship to Other Instruments

Air-Raid Alerting in Ukraine (DBA-ESS-RE-9) and *Repeat Strikes on Rescue Crews in Ukraine* (DBA-ESS-RE-12) read the same campaign through the emergency services. *The Outage of the Bank of England's RTGS System* (DBA-FIN-RE-2) and *The Theft From Bangladesh Bank's Account* (DBA-FIN-RE-1) read the class's two chains in peacetime.

References

National Bank of Ukraine: release on the System of Electronic Payments in 2022; release on Saturday 12 March 2022 as a business day; release on the Power Banking network; release on the new generation of the System of Electronic Payments, 1 April 2023. Deputy Governor of the National Bank of Ukraine, *How National Bank managed to ensure routine operation during Russia's full-scale war*, Kyiv Independent, 13 May 2022. European Payments Council, *Payments in wartime: the story of the National Bank of Ukraine*, 28 July 2023. International Criminal

Court, announcement of two warrants of arrest in the Situation in Ukraine, 5 March 2024. United Nations Human Rights Monitoring Mission in Ukraine, release of 19 September 2024 on attacks on Ukraine's electricity infrastructure. The White House and the United Kingdom government, statements of 18 February 2022 on the denial-of-service attacks on Ukrainian banks. The DBA instruments named in the text, with the Foundational Definitions and FD-REG, and the DBA-MA Parent Framework.

Revision History

Version 0.1, 14 October 2026: issued in Project 11 under the Financial Services Sector Framework v0.5, on the conferral of the framework's §9.1 of 14 October 2026, read against *Payment and Settlement* v0.1. Written new under the Reference Event Compression Standard v0.3, bounding a continuing condition from 24 February 2022 to 28 July 2023. Tier determination at Section 5.2, under FD-DBL §6.4 as ruled on 10 and 13 October 2026; F-1 and F-2, R-1 and R-2. The class's account holds, with two qualifications going to the class as Erratum 2 of Project 11's note of 14 October 2026 (Financial Services batch).

Version 0.2, 14 October 2026: closes OI-4 on the author's ruling of 14 October 2026 that the four characteristics may be read against the campaign on a class's dependencies rather than against the facility as an aim. The tier determination of Section 5.2 stands as written. Nothing else changed. Version 0.1 is in the series' _archive.